

Tecnológico de Monterrey

Campus Santa Fe

SRS

Healthy Growth and Development

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20/9/2026

Introduction

The Institute for Social Security and Services for State Workers (ISSSTE) operates Child Welfare and Development Centers (Estancias para el Bienestar y Desarrollo Infantil, EBDI), whose purpose is to provide protection and promote the development of children from 60 days of age to six years. In this context, ISSSTE seeks to create a platform for registering and consulting the child population enrolled in the various EBDI centers, called “Padrón Infantil”.

The platform will centralize and periodically update information related to each child’s growth and development by recording metrics such as mid-parental target height, weight, height, age, and head circumference. Based on this data, the goal is to monitor development by evaluating nutritional parameters and enabling information to be consulted by individual, EBDI center, state, and region.

Problem Statement

ISSSTE operates Child Welfare and Development Centers (EBDI) across different regions of the country, some owned by the institute and others contracted. Currently, each EBDI’s information is managed in a no structured way, with no centralized registry that brings it together in one place.

This issue means that nutritional assessments do not account for differences in height across the country’s regions. For example, children from northern regions tend to be naturally taller than those from the south. However, because regionalized standards are not applied, their measurements are compared against references that do not correspond to their region, which can lead to them being diagnosed as overweight when they are actually at an appropriate weight for their height and area.

Therefore, a registration and consultation platform is needed that centralizes the information from all EBDI centers in one place and evaluates nutritional status based on each region’s anthropometric specifications, so that the results reflect each child’s actual circumstances.

User Stories

No.	User story
1	As a center user, I want to register a child’s data in the system to determine whether they are healthy.
2	As a center user, I want to record new data every 45 days to observe changes over time.
3	As a center user, I want to access the data of the children registered at my center.
4	As a center user, I want to access charts generated from the stored information to take actions and make decisions regarding the children’s health.
5	As a center user, I want to filter the information in the tables and charts to view a specific item.
6	As a parent, I want to access my child’s data in table format to know the results of their measurements.
7	As a parent, I want to access charts showing my child’s information to understand whether their measurements are healthy.
8	As a parent, I want to filter the information in the charts and tables to view a specific item.

9	As a regional coordinator, I want to access the information of all the states under my jurisdiction to make decisions based on the data presented to me.
10	As a regional coordinator, I want to access the information of the centers within the states under my jurisdiction.
11	As a regional coordinator, I want to access the information of the children within the centers to make decisions.
12	As a regional coordinator, I want to filter the information by state.
13	As a regional coordinator, I want to filter the information by center.
14	As a regional coordinator, I want to filter the information by child.
15	As an administrator, I want to view the data of all children to perform statistical analyses.
16	As an administrator, I want the data to be divided into North, South, East, and West regions.
17	As an administrator, I want the charts to be generated using the current CDC growth tables.
18	As an administrator, I want there to be Weight-for-Height, Height-for-Age, and BMI charts for each child.
19	As an administrator, I want there to be charts by center.
20	As an administrator, I want there to be charts by state.
21	As an administrator, I want there to be charts by region.
22	As an administrator, I want to have user management permissions.

Functional Requirements

ID	Functional requirement
FR01	The system allows the administrator to view data across all Regions, States, EBDIs, and individual profiles.
FR02	The system allows the administrator to filter data by Region, State, EBDI, and individual profiles.
FR03	The system allows the administrator to edit data.
FR04	The system allows the administrator to create and delete users.
FR05	The system allows the administrator to set chart parameters.
FR06	The system allows the regional coordinator to view data for their Region, the States within their Region, the EBDIs within those states, and the profiles for those EBDIs.
FR07	The system allows the regional coordinator to filter data by the States within their Region, the EBDIs within those states, and the profiles for those EBDIs.
FR08	The system allows the facility coordinator to view data for the profiles within their facility.
FR09	The system allows the facility coordinator to filter data by profile ID/registration numbers.

FR10	The system allows the facility coordinator to enter children's data.
FR11	The system allows parents to view their children's data in tabular format.
FR12	The system allows parents to view their children's data in graphical format.
FR13	The system generates a unique ID number for each child.
FR14	The system allows a parent to register using the child's ID number.
FR15	The system features a login page that requires an ID number and password.
FR16	The system divides the total user base into 4 regions: North, South, East, and West.
FR17	The system divides regions into states.
FR18	The system divides states into distinct EBDIs.
FR19	The system features a search bar to search for specific data or specific ID numbers.
FR20	The system includes filters.
FR21	The system stores all user data in a table.
FR22	The system generates charts using user data by region.
FR23	The system generates charts using user data by state.
FR24	The system generates charts using user data by EBDI.
FR25	The system features a profile section displaying the ID number and other data.
FR26	The system automatically calculates nutritional status based on the provided parameters.
FR27	The system calculates age in years and months based on the date of birth.
FR28	The system allows enrolling/registering a child with their general data.
FR29	The system stores the medical history for each individual.
FR30	The system allows unregistering/discharging children when they leave the EBDI.
FR31	The system generates a unique ID number for parents.
FR32	The system allows registering multiple children under the same parent.
FR33	The system has a database where information is stored.
FR34	The system communicates with this database to display information on the website.

Non Functional Requirements

ID	Non-functional requirement
NFR01	The system must store all passwords using a secure hashing algorithm (Argon2)

NFR02	The system must automatically log users out after 30 minutes, asking to log in again (JSONWebToken).
NFR03	Communication with the database always in the backend. Never in the front end.
NFR04	The system must restrict access to information according to each user's profile.

Restrictions

ID	Restriction
R01	The system only accepts the following data: Weight, Height, Mid-Parental Target Height, Age, and Head Circumference.
R02	The system only allows data to be updated every 45 days, unless the administrator modifies it.
R03:	The reference tables are the current CDC tables.
R04:	There are exactly four user profiles.
R05:	Parents can only view their own child's information.

Risk Matrix

1 - 4 Low **5 - 11 Medium** **12 + High**

Risk	Probability	Impact	Level	Mitigation	Contingency
Unauthorized access to children's data due to weak security or roles having unwanted permissions	2 -> Medium	3 -> Medium	**6 -> Medium**	Make sure the roles are clear and that the access is entirely based upon these. Hash passwords and limited sessions	Withdraw the breached credentials immediately and notify the ISSSTE administrator or EBDI manager.
Data entry errors by center staff	4->High	3->Medium	**12->High**	Confirmation screen before submitting data.	Center Staff requests a correction from the administrator, who edits the record with the correct data.
Inconsistent or ambiguous requirements	4->High	3->Medium	**12->High**	Review requirements with the client before development begins.	Hold a meeting with the client to clarify, and update the requirements document.

Usage of the app has resistance from the EBDI and parents due to lacking digital skills	2->Medium	2->Medium	**4->Low**	Perform the accessibility analysis with multiple tools. Also provide simple UI or UI similar to other popular applications.	Provide help from technical staff to offer training on the usage of the application to the users.
Misinterpretation of charts by users	2->Medium	1->Low	**2->Low**	Do usability testing, in order to give the correct explanations and descriptions to the charts.	Provide a user guide and explanatory texts based on feedback.
Loss of history when a child transfers between centers	2->Medium	2->Medium	**4->Low**	Allow transferring a child's record between EBDIs instead of creating a new one.	The administrator manually links the old record to the new center.
Centers miss the 45 day measurement schedule	3->Medium	3->Medium	**9->Medium**	Automatic remainders when a child's next measurement is due.	Generate a report of overdue measurements for regional coordinators.
Legal issues and privacy concerns with Mexican personal data protection law	1->Low	4->High	**4->Low**	Include privacy terms and consent. Also encrypt data in transit and at rest (db)	Cancel data transit and start reviewing with ISSSTE's legal team the measures needed to correct the problematic.